MATHEMATICS Class XI - 2014

(English Version)

Group – A

1. Answer the following questions. (1X10 = 10)

(a) If $A = \{x : 0 < x < 4\}$ and $B = \{x : 3 \le x < 6\}$, where x is an integer, then which of the following is $A \cap B$?

(i) $\{2\}$

(ii) $\{3\}$

(iii) 3

(iv) ϕ

(b) If the $n\ th$ term of an AP is $2n-4$, then which of the following is the common difference?

(i) -2

(ii) $-\frac{1}{2}$

(iii) $\frac{1}{2}$

(iv) 2

(c) Value of $i^n + i^{n+1} + i^{n+2} + i^{n+3}$ is:

(i) 1

(ii) -1

(iii) 0

(iv) None of these.

(d) In a GP, $t_5 : t_3 = 7 : 9$, then $t_9 : t_5$ is:

(i) $7 : 9$

(ii) $9 : 7$

(iii) $81 : 49$

(iv) $49 : 81$

(e) The equation of the circle having centre at (3, 7) and radius 5 is:

(i) $x^2 + y^2 - 6x - 14y + 33 = 0$

(ii) $x^2 + y^2 - 6x - 14y = 33$

(iii) $x^2 + y^2 + 6x + 14y = 33$

(iv) $x^2 + y^2 + 6x + 14y + 33 = 0$

(f) The distance between z-axis and (3, 4, 6) is:

(i) $5\ unit$

(ii) $6 unit$

(iii) $7\ unit$

(iv) None of these.

(g) If $y=\sin^2\frac{x}{2}$, then $\frac{dy}{dx}$ is:

(i) $\sin x$

(ii) $\frac{1}{2}\sin x$

(iii) $\cos x$

(iv) $\frac{1}{2}\cos x$

(h) If $\lim_{x\to 3}\frac{x^n-3^n}{x-3}=27n$, then value of n is:

(i) 3
(ii) 2
(iii) 4
(iv) 5

(i) A and B be two mutually exclusive events such that $P(A)=\frac{3}{8}$, $P(B)=\frac{1}{3}$, then $P(A\cup B)'$ is given by:

(i) $\frac{17}{24}$

(ii) $\frac{2}{9}$

(iii) $\frac{7}{24}$

(iv) $\frac{13}{24}$

(j) The range of 62, 72, 44, 25, 54, 9, 56, 71, 27, -13, -3 is:
(i) 82
(ii) 75
(iii) 85
(iv) 81

<u>Group – B</u>

2.

(a) Answer any two questions: (2X2 = 4)

(i) If $A=\{x:x=3n,\ n\in Z\}$ and $B=\{x:x=6n,\ n\in Z\}$, then find $A\cap B$ and $A\cup B$.

(ii) $A=\{1,2,3,5\}$ and $B=\{4,6,9\}$. A relation R is defined from A to B by $R=\{(x,y):\ the\ difference\ between\ x\ and\ y\ is\ odd\}$. Write R in Roster form.

(iii) Prove that $\tan 54^0=\frac{\cos 9^0+\sin 9^0}{\cos 9^0-\sin 9^0}$.

(iv) If $\tan 15^0=x$, then show that $x^2+2\sqrt{3}x-1=0$.

(b) Answer any one questions: (2X2 = 4)

(i) If ω is an imaginary cube root of unity, then prove that $\dfrac{x\omega^2+y\omega+z}{x\omega+y+z\omega^2}=\left(\dfrac{x\omega+y+z\omega^2}{x\omega^2+y\omega+z}\right)^2$.

(ii) If number of triangles joining angular points of a n-sided polygon be $12n$, find n.

(iii) Find the term independent of x in the expansion of $\left(x-\dfrac{2}{x^2}\right)^{15}$.

(iv) Find the sum, if its exist of the GP: $\dfrac{1}{3}-\dfrac{2}{9}+\dfrac{4}{27}-\dfrac{8}{81}+\ldots$.

(c) Answer any one question: (2X1 = 2)

(i) Find the area of the triangle formed by the straight line $2x-3y=6$ with the co-ordinate axes.

(ii) Find the co-ordinates of the point which divides the line segment joining (2, 3, -8) and (1, -1, 0) internally in the ratio 2:1.

(d) Answer any one question: (2X1 = 2)

(i) Define derivative of $f(x)$ at $x=c$.

(ii) If $\lim\limits_{x\to a} f(x)=\lim\limits_{x\to a} g(x)$, then whether $f(x)=g(x)$ is always true? Justify your answer by an example.

(e) Answer any one question: (2X1 = 2)

(i) If $P(A)=\dfrac{2}{3}$, $P(B)=\dfrac{1}{2}$, $P(A\cap B)=\dfrac{1}{6}$, then find the value of $P(A\cap B')$ and $P(A\cup B)$.

(ii) Find the mean deviation about mean for the following data:
39, 51, 59, 62, 74.

<u>Group – C</u>

3.

(a) Answer any two questions: (4X2 = 8)

(i) For any three sets A,B,C, prove that $A-(B\cup C)=(A-B)\cap(A-C)$.

(ii) If α and β are positive acute angles and $\cos 2\alpha=\dfrac{3\cos 2\beta-1}{3-\cos 2\beta}$, then prove that $\tan\alpha=\sqrt{2}\tan\beta$.

(iii) Solve $\cos^3 x\sin 3x+\sin^3 x\cos 3x=\dfrac{3}{4}$.

(b) Answer any two questions: (4X2 = 8)

(i) Prove by method of principle of mathematical induction:

$$1^2+2^2+3^2+...+(n+1)^2=\frac{(n+1)(+2)(2n+3)}{6}.$$

(ii) If $z=x+iy$ and $|2z+1|=|z-2i|$, then prove that $3(x^2+y^2)+4(x+y)=3$.

(iii) Out of 14 articles, 10 are of same type and each of the remaining is of different types. Find the number of combinations, if 10 articles are taken at a time.

(iv) In the expansion of $\left(\sqrt[3]{2}+\frac{1}{\sqrt[3]{3}}\right)^n$, the ratio of the 7th term from beginning to the 7th term from the end is 1:6, find n.

(v) If the sum of first n terms of a GP is p, sum of its first $2n$ terms is $3p$, prove that the sum of its first $3n$ terms is $7p$.

(c) Answer any two questions: (4X2 = 8)

(i) The equation of two sides of a square is $5x+12y-10=0$ and $5x+12y+29=0$ and the third side passes through (3, 5), find the equation of all other possible sides of the square.

(ii) If the straight line $\frac{x}{a}+\frac{y}{b}=1$ is parallel to the line $4x+3y=6$ and passes through the point of intersection of the lines $2x-y-1=0$ and $3x-4y+6=0$, find values of a and b.

(iii) Find the equation of the circles which passes through the origin and cut off equal chords of length $\sqrt{2}$ units on the straight lines $y=x$ and $y=-x$.

(d) Answer any two questions: (4X1 = 4)

(i) Evaluate $\lim\limits_{x\to 0}\frac{\sin(x^2+4x)}{x^3-5x^2+2x}$.

(ii) Find from the first principle, the derivative of $\tan 2x$ at $x=\frac{\pi}{8}$.

(e) Answer any one questions: (4X1 = 4)

(i) Show that the following statement is true by method of contrapositive.

"If x is an integer and x^2 is even, then x is also even."

(ii) Prove the following by contradiction.

"The sum of a rational and an irrational number is an irrational number."

(f) Answer any one questions: (4X1 = 4)

(i) What is the probability that a year selected at random, in between 2001 and 2010 both inclusive will contain 53 Mondays?

(ii) For a group of 200 students the mean and SD of marks obtained by them were found to be 40 and 15 respectively. Later on, it was found that the score 23 was misread as 32. Find the correct mean and SD.

Group – D

4.

(a) Answer any one questions: (5X1 = 5)

(i) Prove that $\cos^2\alpha + \cos^2(120^0 - \alpha) + \cos^2(120^0 + \alpha) = \frac{3}{2}$.

(ii) Prove that $\tan 70^0 = \tan 20^0 + 2\tan 50^0$.

(b) Answer any two questions: (5X2 = 10)

(i) Draw the graph of the following inequations, and hence obtain the solution region of $3x + 4y \le 12, \quad x \ge -2, \quad y \le 3$, (use graph paper).

(ii) State the Fundamental theorem of Algebra and solve the equation $3x^2 - 5ix + 3 = 0. \quad (i = \sqrt{-1})$

(iii) How many numbers lying between 100 and 1000 can be formed with the digits 0, 3, 4, 6, 8, 9?

(iv) If $S_1, S_2, ..., S_m$ denote the sums of n terms of m numbers of A.P.'s whose first terms are $1, 2, ..., m$ and common differences are $1, 3, ..., (2m-1)$ respectively, show that $S_1 + S_2 + ... + S_m = \frac{mn}{2}(mn+1)$.

(c) Answer any one questions: (5X1 = 5)

(i) Prove that the least focal chord of a parabola is its latus rectum.

(ii) Find the equation of the hyperbola having coordinates of its vertices $(\pm 4, 0)$ and coordinates of its foci is $(\pm 6, 0)$.

(iii) Find the equation of ellipse whose eccentricity is $\frac{1}{2}$, the co-ordinate of one of the focus is (2, 0) and equation of the corresponding directrix is $x - 8 = 0$. Also find out the distance of this focus from its nearest vertex.